

Muhammad Ayaz

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OBJECTIVE

Enthusiastic researcher with a strong background in Computer Vision, Multimodal AI, and Deep Learning, with research experience spanning fire and smoke understanding, vision-language models, affective computing, medical AI, and disaster scene analysis. Interested in developing robust and deployable intelligent systems for real-world applications, with strong proficiency in Python, , PyTorch, TensorFlow, and Keras.

PROFESSIONAL EXPERIENCE

Research Assistant *Mar 2025 - Present*
[Graphics Realization Lab \(GRLab\)](#), [Chung-Ang University](#) *Seoul, South Korea*

- Conduct research in computer vision, image segmentation, multimodal learning, and deep learning under Prof. Sanghyun Seo.
- Develop robust models for fire and smoke understanding, medical vision-language reasoning, and real-world visual perception.

Research Assistant *Sep 2022 - Dec 2024*
Digital Image Processing Laboratory, [Islamia College University](#) *Peshawar, Pakistan*

- Contributed to image classification, 2D and 3D object detection, image segmentation, autonomous driving, and video understanding projects.
- Supported dataset preparation, model training, experimental evaluation, and research documentation.

Course Assistant *May 2023 - Dec 2024*
Digital Image Processing Laboratory, [Islamia College University](#) *Peshawar, Pakistan*

- Assisted with courses in artificial intelligence, computer vision, machine learning, and deep learning.

Junior Researcher *Dec 2020 - 2022*
[National Center for Big Data and Cloud Computing \(NCBC\)](#) *Peshawar, Pakistan*

- Participated in data collection, annotation, preprocessing, and computer vision research tasks.

SELECTED PROJECTS

[ROAD-R Challenge for NeurIPS 2023](#) *2023*

- Worked on autonomous-driving event understanding using the ROAD-R dataset and logical requirements.

[MedVLM: Vision-Language Model for Holistic Medical Applications](#) *2024 - 2025*

- Trained and evaluated multimodal models for medical visual question answering using X-rays, CT scans, and MRIs.
- Collaborated with researchers at [MBZUAI](#) and [United Arab Emirates University](#).

[Cone Detection for Formula Student Cars](#)

- Developed and fine-tuned object detection models for cone-based vehicle navigation using the FSOCCO dataset.

[Fire and Smoke Instance Segmentation in Challenging Environments](#)

- Developed vision models for accurate fire and smoke instance segmentation across indoor and outdoor scenes.
- Improved robustness under difficult lighting, occlusion, scale variation, and complex environmental conditions.

[Anomaly Detection in Video Sequences](#)

- Combined convolutional feature extraction with recurrent temporal modeling to identify abnormal video events.

EDUCATION

MS, Applied Arts and Engineering *2025 - Present*
[Chung-Ang University \(CAU\)](#) *Seoul, South Korea*

- GPA: 4.33/4.5 (98%); current laboratory: [Graphics Realization Lab \(GRLab\)](#).

BS Computer Science *2019 - 2023*
[The University of Agriculture Peshawar \(AUP\)](#) *Peshawar, Pakistan*

- CGPA: 3.30/4.0 (75.91%).
- Thesis: “Toward Reliable Fire & Smoke Instance Segmentation in Challenging Environments: A Vision-Based Approach.”

SELECTED PUBLICATIONS

1. [BoXFire: Geometry-Aware Box-Supervised Fusion for Accurate Fire and Smoke Segmentation](#). *Advanced Engineering Informatics*, 2026. **First author**.
2. [Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey](#). *Computer Methods and Programs in Biomedicine*, 285, 109541, 2026. Co-author.
3. [Privacy-Preserving Action Recognition: Taxonomy, Methods, and Privacy-Utility Trade-offs](#). arXiv:2608.04501, 2026. Co-author.
4. [MedVLM: Vision-Language Model for Holistic Medical Applications](#). *IEEE Consumer Electronics Magazine*, 2025. **First author**.
5. **Supervised Box Fire and Smoke Segmentation in Real-World Scenarios**. 21st International Conference on Multimedia Information Technology and Applications (MITA), 2025. **Best Paper Award**; [press coverage](#).
6. **WVES: A Weighted Ensemble Framework for Student Mental Health Assessment**. 13th International Conference on Technical Education. Accepted.

RESEARCH COLLABORATIONS

Dr. Salman Khan, Research Fellow, [Visual Artificial Intelligence Laboratory](#), Oxford Brookes University, UK

- Collaborated on cone detection, road-crack annotation, fire detection, and introductory 3D object detection using the KITTI dataset.

Dr. Mustaqeem Khan, Assistant Professor, [United Arab Emirates University](#), UAE

- Collaborated on medical vision-language modeling and multimodal violence-detection experiments.

Dr. Alice Othmani, Postdoctoral Researcher, [Ecole Normale Supérieure de Paris](#), France

- Collaborated on [Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey](#).
- Research interests include pattern recognition, image and mesh analysis, contour processing, and shape-feature extraction ([homepage](#)).

ACHIEVEMENTS & SCHOLARSHIPS

- **Best Paper Award**, 21st MITA International Conference on Multimedia Information Technology and Applications, 2025 ([press coverage](#)).
- **Young Scientist Scholarship**, [Chung-Ang University](#), South Korea, 2025 - Present.
- Fully funded [Higher Education Commission](#) scholarship, Pakistan, 2019 - 2023.
- Certificate in Artificial Intelligence, specializing in machine learning and deep learning, awarded by Pakistan's National Vocational and Technical Training Commission.

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, C++, C#, SQL
Python Libraries	OpenCV, Scikit-learn, Scikit-image, NumPy, Matplotlib
Deep Learning Frameworks	PyTorch, TensorFlow, Keras
Research Areas	Computer vision, multimodal AI, vision-language models, segmentation, medical AI

REFERENCES

Dr. Salman Khan

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Dr. Mustaqeem Khan

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